MILITARY SPECIFICATION

GASES, CALIBRATING, AND MIXTURES; GENERAL SPECIFICATION FOR

1. SCOPE

1.1 Scope. This specification covers general requirements for calibrating and mixed gases and cylinder preparation.

1.2 Classification. The gas shall be of the following grades:

Grade A	Primary Standard
Grade B	Operational Standard Grade
Grade C	Gas Mixtures

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

SPECIFICATIONS

MIL-T-27730	Tape, Antiseize, Tetrafluoroethylene, with Dispenser
MIL-V-2	Valves, Cylinder, Gas
MIL-O-27210	Oxygen, Aviator's Breathing, Liquid and Gas
MIL-P-27401	Propellant Pressurizing Agent, Nitrogen
MIL-P-25508	Propellant Oxygen
MIL-A-18455	Argon Technical
BB-H-1168	Helium Technical
BB-N-411	Nitrogen Technical
BB-O-925	Oxygen, Technical, Gas and Liquid

STANDARDS

MIL-STD-101	Color Code for Pipelines and for Compressed Gas Cylinders
MIL-STD-129	Marking for Shipment and Storage
MIL-STD-147	Palletized and Containerized Unit Loads, 40" X 48" Pallets, Skids, Runners, or Pallet-Type Base

(Copies of Military Specifications and Standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

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2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless a specific issue is identified, the issue in effect on date of invitation for bids or request for proposal shall apply.

Department of Transportation (DOT)

49CFR171-190 - Code of Federal Regulations.

(The Code of Federal Regulations (CFR) and the Federal Register (FR) are for sale on a subscription basis by the Superintendent of Document, US Government Printing Office, Wash DC 20402. When indicated, reprints of certain regulations may be obtained from the federal agency responsible for issuance thereof).

American Society for Testing and Materials Publications

ASTM Standards, Parts 30 and 31

(Copies of ASTM publications may be obtained upon application to the American Society for Testing and Materials, 1916 Race Street, Philadelphia PA 19102.)

Technical Society and Technical Association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using federal agencies.

3. REQUIREMENTS

3.1 Materials. Unless otherwise specified (6.2), the minor components shall conform to the general requirements specified in Table I.

3.2 Specification Sheets. Specific gas compositions shall be detailed in the applicable specification sheets.

3.3 Limiting Values. The following applies to all specified limits of this specification. For purposes of determining conformance with these requirements, an observed value or a calculated value shall be rounded off "to the nearest unit" in the last right-hand place of figures used in expressing the limitation value, in accordance with the rounding-off method of the Recommended Practices for Designating Significant Places in Specified Limiting Values (ASTM Designation: #29-67).

3.4 Cylinder Maintenance. Unless otherwise specified (6.2), each Government- and supplier-furnished cylinders shall be processed by the supplier in accordance with section 5.

3.5 Capacity. Unless otherwise specified (6.2), cylinders shall be filled to within 200 psig of rated pressure.

3.6 Leakage. Cylinders and valves shall not leak after being filled.

3.7 Rejected Cylinders. Government-owned cylinders considered unsatisfactory for filling or refilling shall be processed as specified in the contract or purchase order (6.2). Other cylinders shall be disposed of by the contractor.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for Inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except

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as otherwise specified in the contract or order, the supplier may use his own or other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements (6.2).

4.1.1 Points of Inspection (6.2)

4.1.1.1 Cylinders. Unless otherwise specified, acceptance tests of calibrating and mixed gases in cylinders shall be conducted at the site of filling. The acceptance tests shall be completed prior to shipment of cylinders from the shipping site.

TABLE I

GENERAL REQUIREMENTS

	Preparation Method	Preparation Tolerance	Analytical Accuracy
Grade A	Gravimetrically on high-load high-sensitivity balance	± 5% of each minor component	± 0.02 % absolute or ± 1% relative for each min or component (whichever is smaller)
Grade B	Most accurate means available other than by weight	± 10% of each minor component greater than 50 ppm ± 20% of each minor component less than 50 ppm	± 2% of each minor component greater than 50 ppm ± 5% of each minor component less than 50 ppm
Grade C	Most accurate means available other than by weight	± 10% of each minor component greater than 50 ppm ± 20% of each minor component less than 50 ppm	Testing not required

4.2 Classification of Tests. The inspection and testing of the product shall be classified as quality conformance tests.

4.3 Quality Conformance Tests. Quality conformance tests shall consist of:

4.3.1 Grades A and B

4.3.1.1 Individual Tests 4.3.3

4.3.2 Grade C

4.3.2.1 Individual Tests 4.3.4

4.3.3 Individual Tests. Each cylinder shall be subjected to the following test as described under 4.6, 4.7, and applicable specification sheet.

Product Conformance Tests ..... 4.6 & Specification Sheet

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Leakage Test 4.7.2

Filling Pressure Test 4.7.1

4.3.4 Individual Tests. Each cylinder shall be subjected to the following test as described under 4.7:

Leakage test 4.7.2

Filling Pressure Test 4.7.1

4.3.5 Sample. Each sample shall be of sufficient size to conduct all the quality conformance tests as specified herein. Unless otherwise specified, the quality conformance tests shall be made on each required sample (6.2). When required, an equivalent sample shall be forwarded to a laboratory designated by the procuring activity for subjection to the quality conformance tests specified herein.

4.3.6 Sampling. A sample shall be taken from each cylinder filled with calibrating gas.

4.4 Rejection. When any samples of the product tested as specified herein fail to conform to the requirements specified herein, the cylinder represented by the sample shall be rejected.

4.5 Test Conditions. Unless otherwise specified (6.2), the test conditions herein specified shall apply to the test method in 4.6.

4.5.1 Chemicals. All chemical tests shall be made with ACS specification reagent chemicals and distilled water.

4.5.2 Calibration. The accuracy of measurement equipment used in preparing and analyzing these standards shall be traceable to the National Bureau of Standards.

4.6 Test Methods.

4.6.1 Water Content shall be determined by any of the following procedures:

4.6.1.1 By an electrolytic hygrometer having an indicator graduated in ppm (v/v) on a range which is no greater than ten times the specified maximum moisture content. The analyzer shall be operated so that the sensitivity for water is either 0.5 ppm or 10% of the specified maximum amount, whichever is greater.

4.6.1.2 By a piezoelectric sorption hygrometer on a range which is no greater than ten times the specified maximum moisture content. The analyzer shall be operated so that the sensitivity for water is either 0.5 ppm or 10% of the specified maximum amount, whichever is greater.

4.6.1.3 By an aluminum oxide capacitor-equipped analyzer on a range which is no greater than ten times the specified maximum moisture content. The analyzer shall be operated so that the sensitivity for water is either 0.5 ppm or 10% of the specified maximum amount, whichever is greater.

4.6.1.4 Any other method approved by the procuring activity.

4.6.2 Total hydrocarbon content shall be determined by any one of the following procedures:

4.6.2.1 By a flame ionization type analyzer. The analyzer is to be calibrated (zeroed and spanned) at appropriate intervals by the use of calibration gas standards. The range used should be no greater than ten times the specified maximum gaseous hydrocarbon content expressed as methane. The analyzer shall be operated so that its sensitivity for methane is either 0.5 ppm or 10% of the specified maximum amount, whichever is greater.

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4.6.2.2 By a gas cell equipped infra-red analyzer. The analyzer is to be calibrated at appropriate intervals with methane in major component gas standards at approximately 3.5 microns (the characteristic absorption wave length for C-H stretching). The analyzer shall be operated so that its sensitivity for methane is either 0.5 ppm or 10% of the specified maximum amount, whichever greater.

4.6.2.3 By a gas chromatograph in accordance with 4.6.5.1.

4.6.2.4 By a mass spectrometer in accordance with 4.6.5.2.

4.6.2.5 Any other method approved by the procuring activity.

4.6.3 Gaseous hydrocarbon content shall be determined by any one of the following procedures:

4.6.3.1 By a gas cell equipped infrared analyzer appropriately calibrated with the specified gaseous hydrocarbon in major component gas. The analyzer shall be operated so that its sensitivity for the specified hydrocarbon is either 0.5 ppm or 10% of the specified maximum amount, whichever is greater.

4.6.3.2 By a gas chromatograph in accordance with 4.6.5.1.

4.6.3.3 By a mass spectrometer in accordance with 4.6.5.2.

4.6.3.4 Any other method approved by the procuring activity.

4.6.4 The oxygen content shall be determined by one of the following procedures:

4.6.4.1 By an electrochemical type oxygen analyzer containing an aqueous electrolyte. The analyzer is to be calibrated at appropriate intervals by the use of calibration gas standards or integrally in accordance with Faraday's Law. The range used should be no greater than ten times the specified maximum oxygen content. The analyzer shall be operated so that its sensitivity for oxygen is either 0.5 ppm or 10% of the specified maximum amount, whichever is greater.

4.6.4.2 By a heat-of-reaction type analyzer. The analyzer is to be calibrated at appropriate intervals by the use of calibrated gas standards or integrally in accordance with Faraday's Law. The range used should be no greater than ten times the specified maximum oxygen content. The analyzer shall be operated so that its sensitivity for oxygen is either 0.5 ppm or 10% of the specified maximum amount, whichever is greater.

4.6.4.3 By an analyzer in which oxygen reacts to form a compound which is subsequently measured. The analyzer is to be calibrated at appropriate intervals by the use of calibration gas standards. The range used should be no greater than ten times the specified maximum oxygen content. The analyzer shall be operated so that its sensitivity for oxygen is either 0.5 ppm or 10% of the specified maximum amount, whichever is greater.

4.6.4.4. By a gas chromatograph in accordance with 4.6.5.1.

4.6.4.5 By a mass spectrometer in accordance with 4.6.5.2.

4.6.4.6 Any other method approved by the procuring activity.

4.6.5 The argon, nitrogen, and neon contents - shall be determined by any one of the following methods:

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4.6.5.1 By a gas chromatograph - this method may be used not only for argon, nitrogen, and neon, but may also be used for the determination of any of the other limiting characteristic gaseous components. The analyzer must be capable of separating and determining the component with a sensitivity of 0.5 ppm or 10% of the specified maximum amount of the component, whichever is greater. Appropriate impurity concentrating techniques may be used to attain the sensitivity. The analyzer is to be calibrated at appropriate intervals by the use of calibration gas standards which contain the applicable limiting characteristic (minor) gaseous components in major component gas.

4.6.5.2 Mass spectrometer. This method may be used not only for argon, nitrogen, and neon, but may also be used for the determination of many of the other minor components in calibrating gases. The analyzer must be capable of separating and determining the minor component with a sensitivity of 0.5 ppm or 10% of the specified maximum amount of the component, whichever is greater. Appropriate impurity concentrating techniques may be used to attain the sensitivity. The analyzer is to be calibrated at appropriate intervals using the appropriate calibrating techniques for a mass spectrometer.

4.6.5.3 Any other method approved by the procuring activity.

4.6.6 The hydrogen content shall be determined by any one of the following procedures:

4.6.6.1 By a heat-of-reaction type analyzer. The analyzer is to be calibrated at appropriate intervals by the use of calibration gas standards or integrally in accordance with Faraday's Law. The range used should be no greater than ten times the specified maximum hydrogen content. The analyzer shall be operated so that its sensitivity for hydrogen is either 0.5 ppm or 10% of the specified maximum amount, whichever is greater.

4.6.6.2 By an analyzer in which hydrogen reacts to form a compound which is subsequently measured. The analyzer is to be calibrated at appropriate intervals by the use of calibration gas standards. The range used should be no greater than ten times the specified maximum hydrogen content. The analyzer shall be operated so that its sensitivity for hydrogen is either 0.5 ppm or 10% of the specified maximum amount, whichever is greater.

4.6.6.3 By a gas chromatograph in accordance with 4.6.5.1.

4.6.6.4 By a mass spectrometer in accordance with 4.6.5.2.

4.6.6.5 Any other method approved by the procuring activity.

4.6.7 The carbon monoxide content shall be determined by one of the following procedures:

4.6.7.1 By a catalytic combustion analyzer. The analyzer is to be calibrated (zeroed and spanned) at appropriate intervals, with calibration gas standards. The range used should be no greater than ten times the specified maximum amount. The analyzer shall be operated so that its sensitivity for carbon monoxide is either 0.5 ppm or 10% of the specified maximum amount whichever is greater.

4.6.7.2 By a gas cell-equipped dispersive or non-dispersive infrared analyzer. The analyzer is to be calibrated (zeroes and spanned) at appropriate intervals with calibration gas standards, at approximately 4.6 microns. The analyzer shall be operated to that its sensitivity for carbon monoxide is either 0.5 ppm or 10% of the specified maximum amount whichever is greater.

4.6.7.3 By a gas chromatograph in accordance with 4.6.5.1.

4.6.7.4 By a mass spectrometer in accordance with 4.6.5.2.

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4.6.7.5 Any other method approved by the procuring activity.

4.6.8 The carbon dioxide content shall be determined by one of the following procedures:

4.6.8.1 By a gas cell-equipped dispersive or non-dispersive infrared analyzer. The analyzer is to be calibrated at appropriate intervals with calibration gas standards at approximately 4.3 microns. The analyzer shall be operated so that its sensitivity for carbon dioxide is 0.5 ppm or 10% of the specified maximum amount, whichever is greater.

4.6.8.2 By a gas chromatograph in accordance with 4.6.5.1.

4.6.8.3 By a mass spectrometer in accordance with 4.6.5.2.

4.6.8.4 Any other method approved by the procuring activity.

4.6.9 The nitrogen dioxide content shall be determined by one of the following procedures:

4.6.9.1 By a gas cell-equipped infrared analyzer. The analyzer is to be calibrated at appropriate intervals with calibration gas standards at approximately 6.2 microns. The range used should be no greater than ten times the specified maximum amount. The analyzer shall be operated so that its sensitivity for nitrogen dioxide is at either 0.5 ppm or 10% of the specified maximum amount whichever is greater.

4.6.9.2 By a gas chromatograph in accordance with 4.6.5.1.

4.6.9.3 By a mass spectrometer in accordance with 4.6.5.2.

4.6.9.4 Any other method approved by the procuring activity.

4.6.10 The nitrous oxide content shall be determined by one of the following procedures:

4.6.10.1 By a gas cell-equipped infrared analyzer. The analyzer is to be calibrated at appropriate intervals with calibration gas standards, at approximately 4.5 microns. The analyzer shall be operated so that its sensitivity for nitrous oxide is at either 0.5 ppm or 10% of the specified maximum amount whichever is greater.

4.6.10.2 By a gas chromatograph in accordance with 4.6.5.1.

4.6.10.3 By a mass spectrometer in accordance with 4.6.5.2.

4.6.10.4 Any other method approved by the procuring activity.

4.6.11 Thy sulfur dioxide content shall be determined by one of the following procedures:

4.6.11.1 By a gas cell-equipped infrared analyzer. The analyzer is to be calibrated at appropriate intervals with calibration gas standards at approximately 7.3 microns. The analyzer shall be operated so that its sensitivity for sulfur dioxide is either 0.5 ppm or 10% of the specified maximum amount, whichever is greater.

4.6.11.2 By a gas chromatograph in accordance with 4.6.5.1.

4.6.11.3 By a mass spectrometer in accordance with 4.6.5.2.

4.6.11.4 Any other method approved by the procuring activity.

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4.6.12 The halogenated solvent or refrigerant content shall be determined by one of the following procedures:

4.6.12.1 By an electronic type halide detector. The analyzer is to be calibrated (zeroed and spanned) at appropriate intervals with gaseous calibration standards. The range used should be no greater than ten times the specified maximum amount. The analyzer shall be operated so that its sensitivity for halogenated solvents or refrigerants is either 0.5 ppm or 10% of the specified maximum amount, whichever is greater.

4.6.12.2 By a gas cell equipped infrared analyzer appropriately calibrated with a halogenated compound. The analyzer should be operated so that its sensitivity for halogenated solvents or refrigerants is either 0.5 ppm or 10% of the specified maximum amount, whichever is greater.

4.6.12.3 By a gas chromatograph in accordance with 4.6.5.1.

4.6.12.4 By a mass spectrometer in accordance with 4.6.5.2.

4.6.12.5 Any other method approved by the procuring activity.

4.6.13 The acetylene content shall be determined by one of the following procedures:

4.6.13.1 By a gas cell equipped infrared analyzer. The analyzer is to be calibrated (zeroed and spanned) at appropriate intervals with calibration gas standards at approximately 13.7 microns. The analyzer shall be operated so that its sensitivity is at either 0.5 ppm or 10% of the specified maximum amount whichever is greater.

4.6.13.2 By a gas chromatograph in accordance with 4.6.5.1.

4.6.13.3 By a mass spectrometer in accordance with 4.6.5.2.

4.6.13.4 Any other method approved by the procuring activity.

4.7 Gas cylinder tests.

4.7.1 Filling pressure. Cylinders shall be tested for proper filling pressure by attaching a calibrated Bourdon tube or equal gauge to the valve outlet and by attaching either a thermocouple or thermometer to the cylinder wall. The gauge shall have scale divisions not greater than 10 psi. If a thermometer is used, tape or putty shall be applied to the bulb to protect it from alien temperatures. Putty shall not be applied between the bulb and the cylinder wall. The thermometer shall have scale divisions not greater than 2°F or 1°C. The cylinder shall be stabilized to approximately room temperature. The cylinder valve shall be opened and the internal pressure observed on the gauge. If the internal pressure (corrected to 70°F) differs from the applicable pressure value by more than 200 psi, the sampled cylinder shall be rejected.

4.7.2 Leakage. Each cylinder shall be tested for leaks at the neck threads, stem packing, and safety device of the valve with leak-detection fluid. Valve seat leakage shall be tested after filling has been completed. This shall be done by applying the leak-detection fluid sparingly across the outlet of the valve. Only leak-detection fluid that leaves no residue shall be used on the outlet. Fluids that give up moisture to the product during discharge of the cylinder shall not be used.

4.8 Preparation for delivery inspection. The packaging and marking for shipment and storage of the calibrating gas shall be inspected to determine compliance with the requirements of section 5 of this specification.

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5. Preparation for Delivery.

5.1 Packaging. The product furnished under this specification is a hazardous material as defined and regulated by the Department of Transportation (DOT) regulations. All packaging to be shipped commercially by any mode of transportation shall comply with the requirements of DOT regulations 49 CFR 171-190, or DOT special permit. The product shall be packaged in containers and unit quantities as specified by the procuring activity (6.2).

5.2 Preparation of containers. The contractor shall establish the condition of all containers and insure that 5.3 as applicable is complied with. Contractor-owned containers shall be cleaned, tested, and repaired by the contractor at his own expense. Leased or government-owned containers shall be cleaned, tested, and repaired in accordance with the schedule established in the contract or purchase order (6.2).

5.3 Gas cylinders. Qualification, maintenance, cleaning, and use of cylinders shall conform to the Title 49 CFR requirements and as follows:

5.3.1 Structural damage. Cylinders shall be inspected for damage such as dents, gouges, deep scratches, arc burns, and fire exposure in excess of DOT allowable limits. Unacceptable cylinders shall be rejected and processed in accordance with 5.3.6.

5.3.2 Painting and markings. Cylinders shall be inspected for condition of paint and name stencil in accordance with 5.5. Paint condition shall preclude metal corrosion due to weather and clearly convey its identification meaning. Name stencils shall be correct for the material inside the cylinder, clearly readable, and size in accordance with MIL-STD-101.

5.3.3 Hydrostatic date. Cylinders shall be inspected for the date of last hydrostatic test. Unless otherwise specified (6.2), cylinders destined for CONUS or overseas with an expiration date within three or six months of five years, respectively, will be hydrostatically tested in accordance with DOT requirements prior to filling. Rejected cylinders shall be processed in accordance with 5.3.6.

5.3.4 Valves. Cylinder valves shall be inspected for proper operation and damage. Any valve found damaged or otherwise unsatisfactory shall be repaired if only minor adjustments are required or replaced with a new valve conforming to MIL-V-2. Valves shall be tightened into the cylinder with a minimum of five threads engaged.

5.3.4.1 Thread tape. When valves are installed or removed and reinstalled, apply thread tape conforming to MIL-T-27730. Wrap the tape under tension starting 3 threads back from the end and in the direction with the male pipe thread helix toward thread runout. Wrap once plus an overlap of 1/2-inch at thread run out end.

5.3.5 Residual Gas Removal. Each cylinder shall be prepared as follows:

5.3.5.1 Grades A and B. The cylinder shall be prepared by attaching a vacuum pump to the cylinder valve outlet, opening the valve, and heating the cylinder in an oven or hot-water bath to approximately 180°F (82.2°C) while evacuating the cylinder to 100 mmHg or less pressure for a minimum of 30 minutes. Each cylinder shall then be charged to 10 psig with major component gas.

5.3.5.2 Grade C. The cylinder shall be prepared by attaching a vacuum pump to the cylinder valve outlet, opening the valve and evacuating the cylinder to 100 mmHg or less. Each cylinder shall then be charged to 10 psig with major component gas.

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5.3.6 Rejected cylinders. Government-owned cylinders considered unsatisfactory for filling shall be processed as specified in the contract or purchase order (6.2). Other cylinders shall be disposed of by the contractor.

5.3.6.1 Rejection tag. A tag bearing the reason for rejection shall be firmly wired to the cylinder valve. The information shall be legible and the tag and markings shall be impervious to climatic conditions.

5.3.7 Filling. Just prior to the filling or mixing operation, the cylinders shall be evacuated to 100 mmHg or less pressure.

5.4 Palletization. When specified (6.2) cylinders shall be palletized in accordance with MIL-STD-147.

5.5 Labeling and marking. Each cylinder shall be labeled and placarded in accordance with MIL-STD-129, and established DOT regulations or special permit. In addition, an identification tag, and cylinder color code shall be used.

5.5.1 Identification tag. Unless otherwise specified in the contract or purchase order, an identification tag impervious to climatic conditions shall be wired to the outlet port of each container and shall contain the following information: Product name, specification number with revision letter, type designation, FSN number, quantity, name of manufacturer, name of contractor (if different from manufacturer), date of manufacture, and lot identification number (6.2).

5.5.2 Cylinder color code. Unless otherwise specified by the procuring acitivity, each cylinder shall be color coded in accordance with MIL-STD-101. Any other name identification shall be obliterated by removing or overpainting.

6. NOTES

6.1 Intended Use.

6.1.1 Grade A is used for calibrating analytical equipment where maximum accuracy is required and as the standard for analyzing secondary or working standards.

6.1.2 Grade B is used for calibrating analytical equipment where extreme accuracy is not required.

6.1.3 Grade C is used in applications which do not require mixtures of accurately known compositions.

6.2 Ordering Data. Purchasers should select the preferred options permitted herein and include the following information in procurement documents:

6.2.1 Procurement Requirements.

6.2.1.1 Title, number, and date of this specification and applicable specification sheet.

6.2.1.2 Method of shipment, type, and capacity of cylinders (3.5, 5.1, and 6.4).

6.2.1.3 When variations to minor components are required (3.1).

6.2.1.4 When variations to cylinder maintenance are granted (3.4 and 5.2).

6.2.1.5 When variations to filling of cylinders to rated capacity are granted (3.5).

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6.2.1.6 When instructions for disposition of rejected cylinders are required (3.7 and 5.3.6).

6.2.1.7 When inspection requirements are to be performed by other than supplier (4.1).

6.2.1.8 When variations in points of inspection are granted (4.1.1).

6.2.1.9 When variation in sampling is granted (4.3.5).

6.2.1.10 When variations in test conditions are granted (4.5).

6.2.1.11 When cleaning and repair schedule is required for leased or government-owned cylinders (5.2).

6.2.1.12 When palletization of cylinders is required (5.4).

6.2.1.13 When identification tag is to be other than as specified (5.5.1).

6.2.1.14 When the cylinder color code is to be waived (3.4.1 and 5.5.2).

6.2.1.15 When extended retest period is granted (5.3.3).

6.2.2 Contract Data Requirement. Data conforming to Data Item Description DI-T-3733 Quality Conformance Test Reports (Fuels), is a requirement for delivery in connection with this specification. The data item will be specified for delivery on the DD Form 1423. The test report for grades A and B should include the cylinder serial number, major component, and each minor component, the specifications sheet number, prepared concentration analysis, and type instrument used for analysis. The test report for Grade C should include the major component, each minor component, the specification sheet, and the prepared concentration.

6.3 Performance of Services. The gas supplier should furnish at no additional cost all services which are required at each and every filling of a cylinder to comply with applicable regulations and normal standard practice. Such services would include, but not be limited to, all inspection, testing, evaluation, and handling required for the gas supplied.

6.3.1 A schedule of allowable fees should be specified by the purchaser for the gas supplier's performance of services such as the replacement of valves, valve parts, and cylinder caps, hydrostatic testing, cleaning, painting, color coding, marking, and handling of unserviceable cylinders as required. All materials and components for these services may be furnished by the gas supplier.

6.4 Basis of Purchase. Calibrating or mixed gas should be purchased by standard cubic feet (NTP) in DOT approved cylinders (6.2).

6.5 Pollution Control. US Public Laws dictate increased effort to improve air, land, and water pollution control of toxic vapor, leaks, spills, and disposal during all phases of manufacture, transfer, storage, and transportation operations. The manufacture/supplier is enjoined to approach the appropriate pollution control districts to mutually resolve all problem areas, and to develop adequate control and disposal methods for situations which are likely to develop in any of the phases.

6.6 Precautions on mixed and calibrating gases:

6.6.1 Components that react cannot be mixed.

6.6.2 When components absorb and desorb with the cylinder wall, mixtures in the trace or factional parts per million range cannot normally be prepared.

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6.6.3 Flammable mixtures of fuels and oxidizers will not be prepared. Normally mixtures of fuels in oxidizers will be limited to 25% of the lower explosive limit (L.E.L).

6.6.4 Mixtures containing a liquefied gas component can possibly condense thereby unmixing the componets.

6.6.5 Moisture reference mixtures. Moisture mixtures are not standards but are only instrument references for field use. Some suppliers use a catalytic conversion of a predetermined gas mixture across a catalyst bed to develop a moisture standard that is independent from the absorption disportion phenomenon which occurs on the cylinder walls due to variance in cylinder pressure and temperature.

6.7 Definitions.

6.7.1 Percent (v/v) - Molar percent.

6.7.2 ppm (v/v) - Molar parts per million.

6.7.3 "as methane" is defined as the single carbon atom equivalent.

6.7.4 Calibration gas standards are gas mixtures of known composition used for the calibration of an instrument.

6.7.5 Sensitivity of an analytical procedure is the minimum value of the limiting characteristic being determined which will produce a positive indication.

6.7.6 Flammable gas is any gas at NTP (1) that forms a flammable mixture with air at a concentration of 13% or less or (2) that has a flammable range wider than 12% with air regardless of the lower flammable limits.

6.7.7 Lower Explosive Limit (LEL) is minimum percent by volume of a gas which when mixed with air at NTP will form a flammable mixture.

6.7.8 Normal Temperature and Pressure (NTP) is defined as 70°F and 14.696 PSIA.

6.7.9 Standard Temperature and Pressure (STP) is defined as 0°C and 760 mm of mercury.

6.7.10 Span gas (test gas) is used to calibrate the analyzer for the quantitative measurement of gas sample. The concentration of the minor component should be in the same order of magnitude as the sample for proper calibration. The major component in both the span gas and sample should be the same.

6.7.11 Zero gas is used to establish the baseline and as a reference point to "zero" an analyzer. ("Zero" is the lower limit of the range being used and not necessary to value zero). It is used to eliminate the effect of background gases on the analyzer. The major component in both the zero gas and the sample should be the same.

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